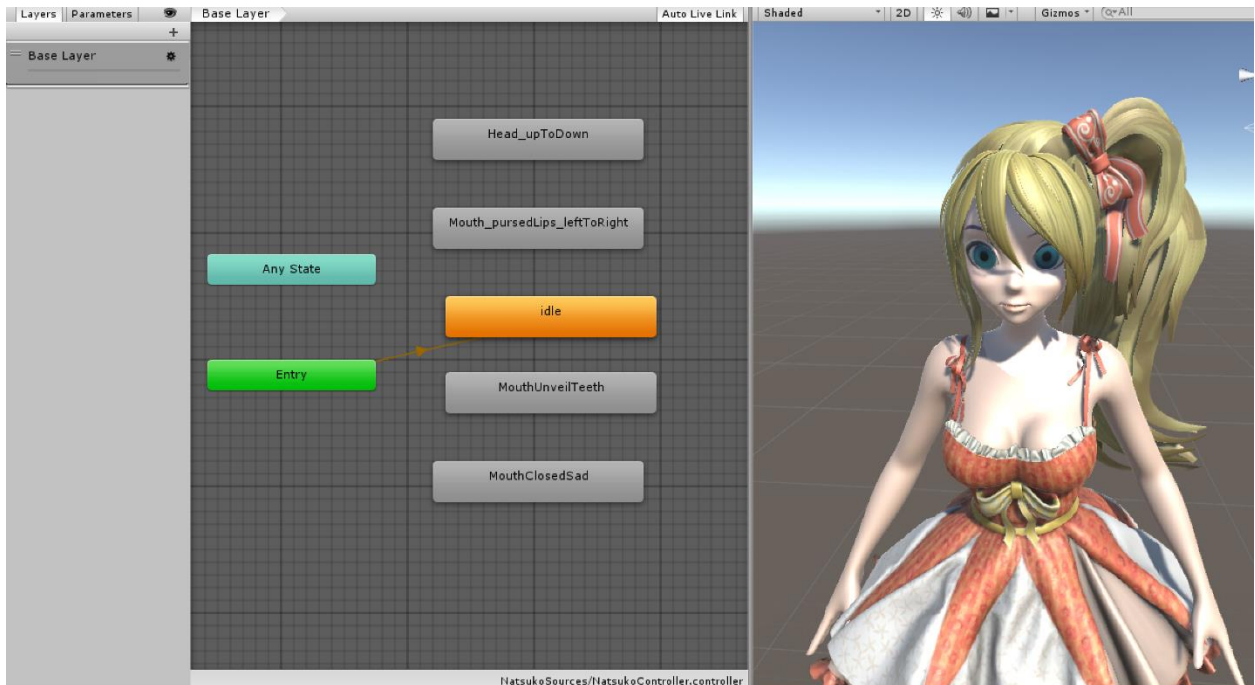


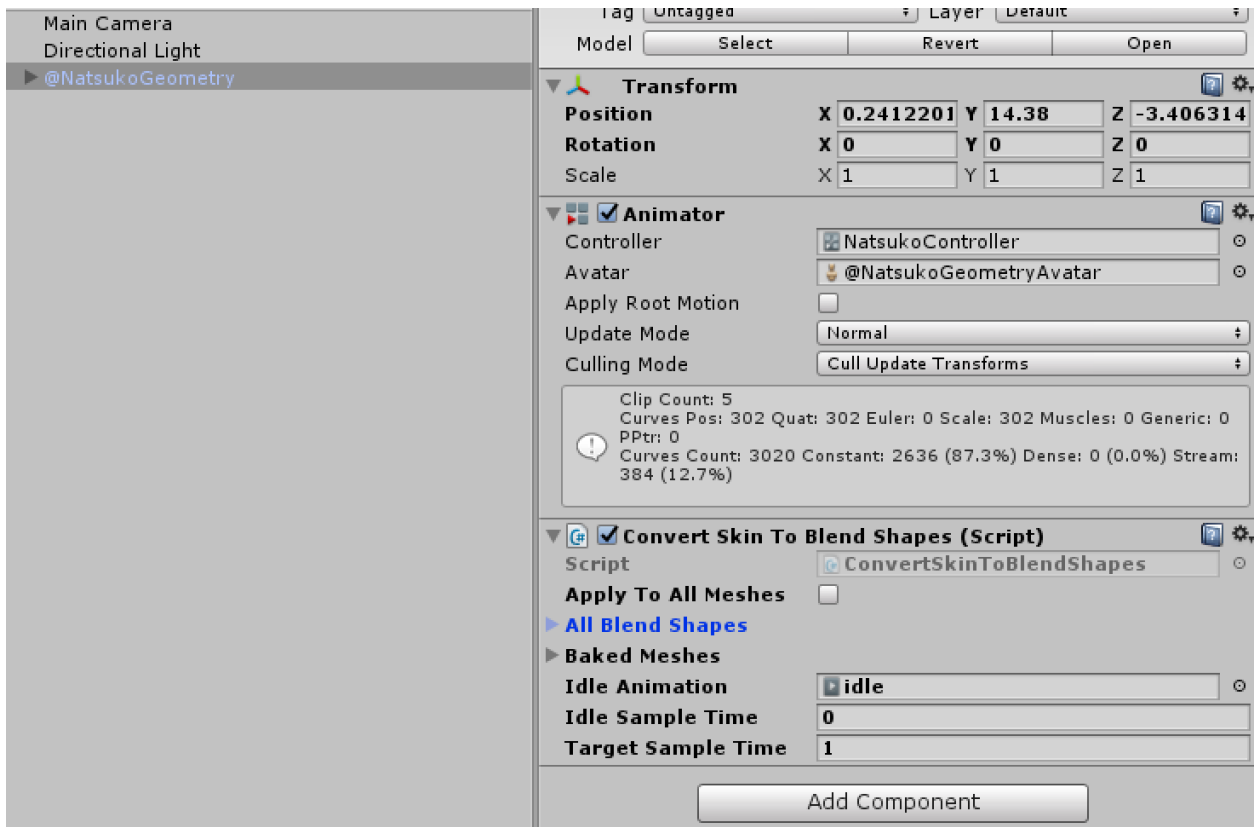
Skinning To BlendShapes

1. Create an animation controller if you don't have one already and add all animations into it. Just drag and drop animation files into the controller's layer. All animations present here will generate a blendshape.

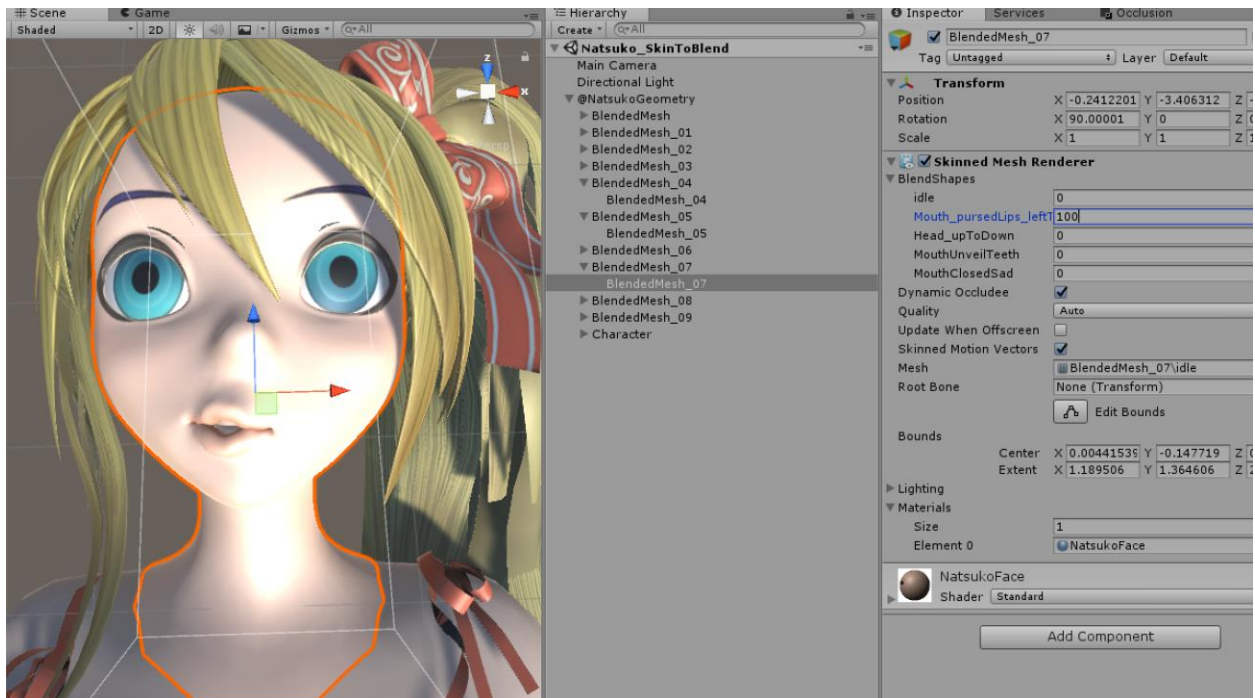


2. Next, apply this animation controller to your character's root game object. In this example "NatsukoController" was selected. Then add a new component and select the "ConvertSkinToBlendShapes" script. The process is fairly simple. Drag and drop your idle animation to be used as the blendshape base or default state. Since an animation can vary over time, you are able to select at what time of the animation this base state should be created from. Choose an adequate "SampleTime" for your idle animation (Idle Sample Time). All other animations will be sampled at "Target Sample Time" and a blendshape from that state will be created. In this example, each animation has an action and the action's maximum

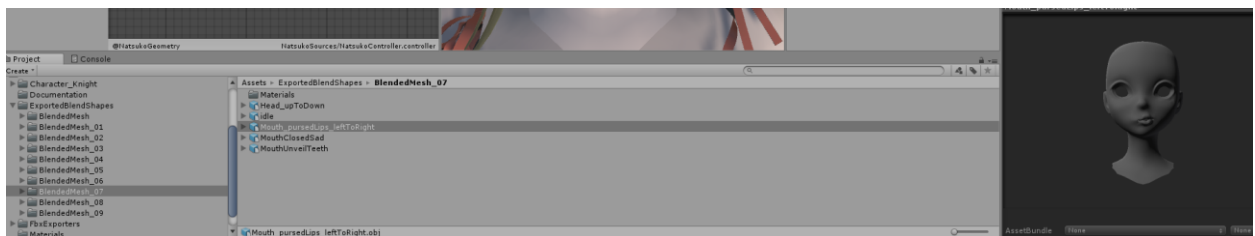
intensity is at 100% of the animation and since all animations are 1 second length, Target Sample Time is set to 1;



3. Next, you should press the Unity Play button. Blendshapes are created at run-time and new meshes are generated that undo skinning transformations (this is to ensure the blendshape works correctly for cases where the skinning default pose is different than the idle animation pose). Only these newly created meshes have blendshapes on. Select one of these newly created meshes (in this example it is "BlendedMesh_07" and you can check in realtime how the blendshape is applied.

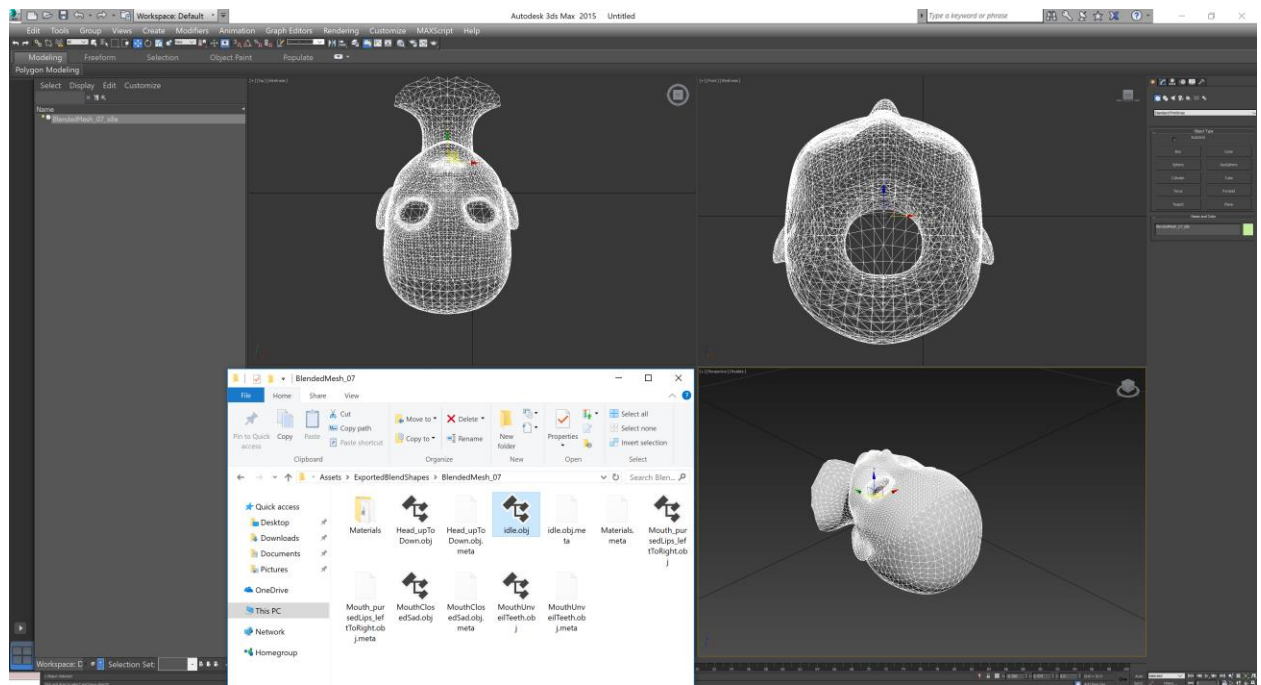


4. Now you will have to refresh Unity's assets, so right click anywhere in the project structure and select "Refresh". After this, you will have a new folder :
 "Assets/ExportedBlendShapes/_ObjectName_" (in this example it is
 "Assets/ExportedBlendShapes/BlendedMesh_07"). Here, you will find an OBJ file for each
 blendshape target, named after the animation used to generate it.

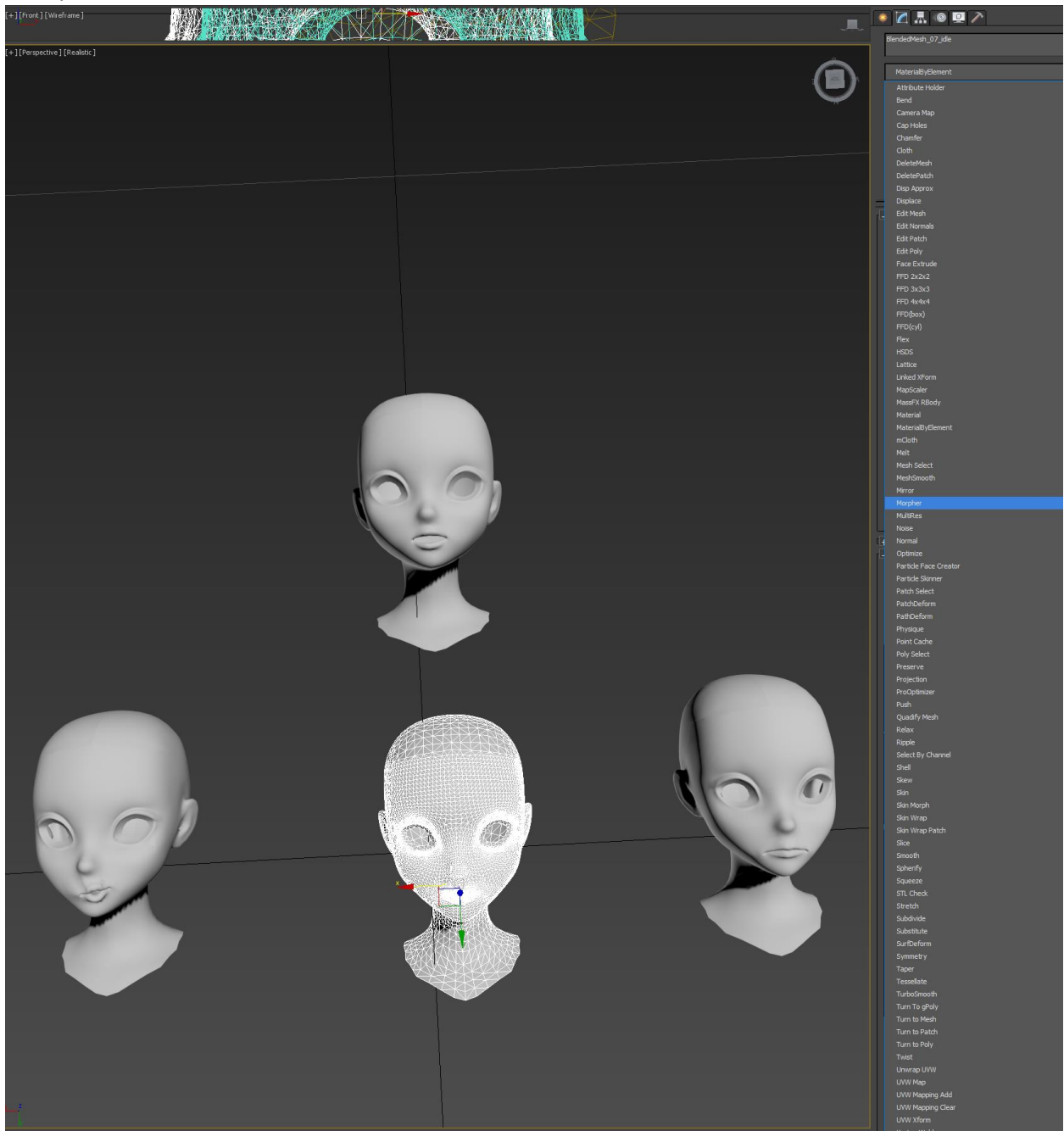


5. This tutorial will show you how to create the blendshapes in 3D Studio Max, but other software like Maya or ZBrush should have a similar workflow. In 3D Studio Max, import or drag and drop each obj file from the folder specified above. Start with idle. Idle is the base of all BlendShapes.

5.

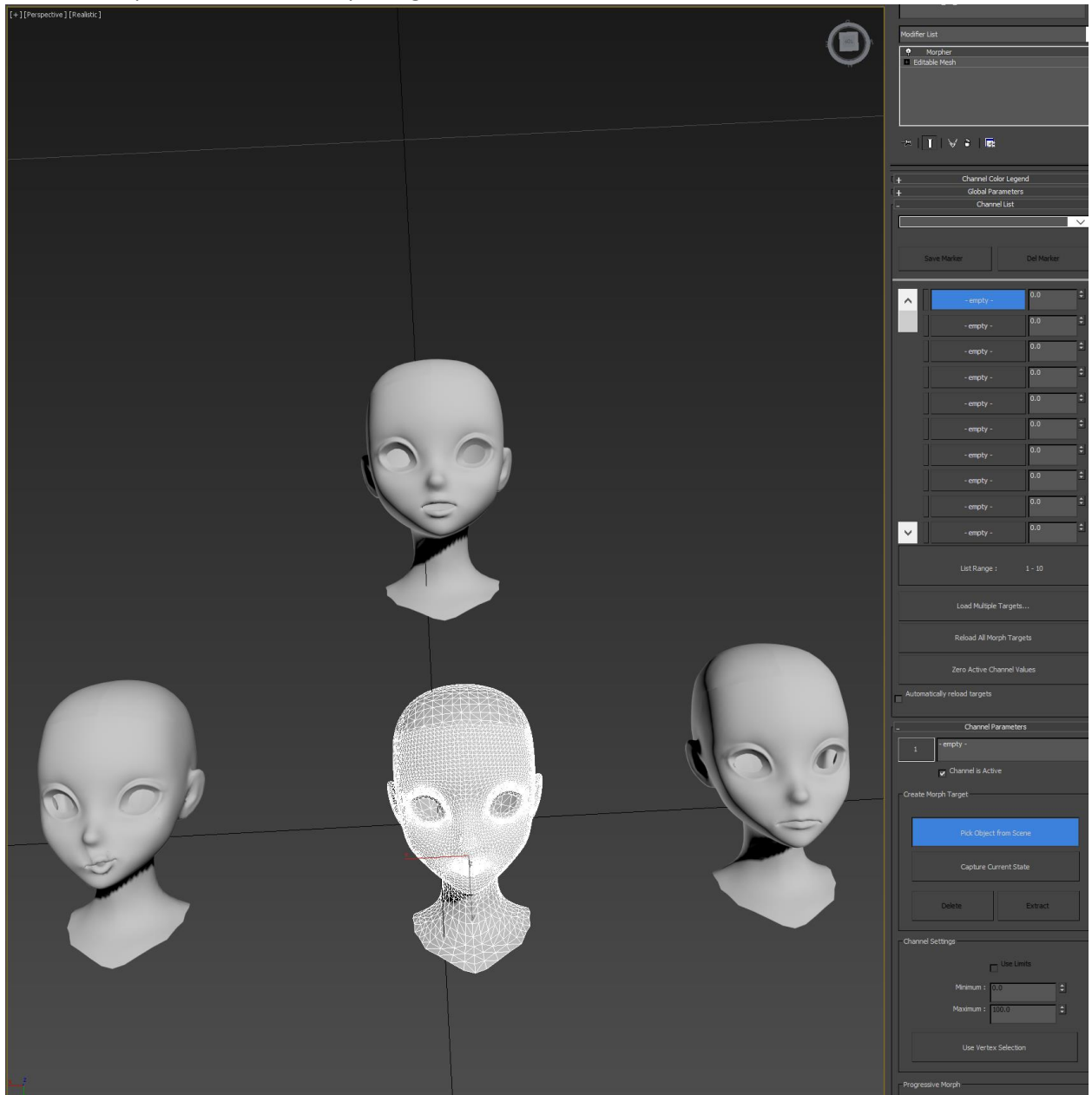


6. After adding all OBJ files in the scene, select the Idle one, go to modifiers list and choose “Morpher”.

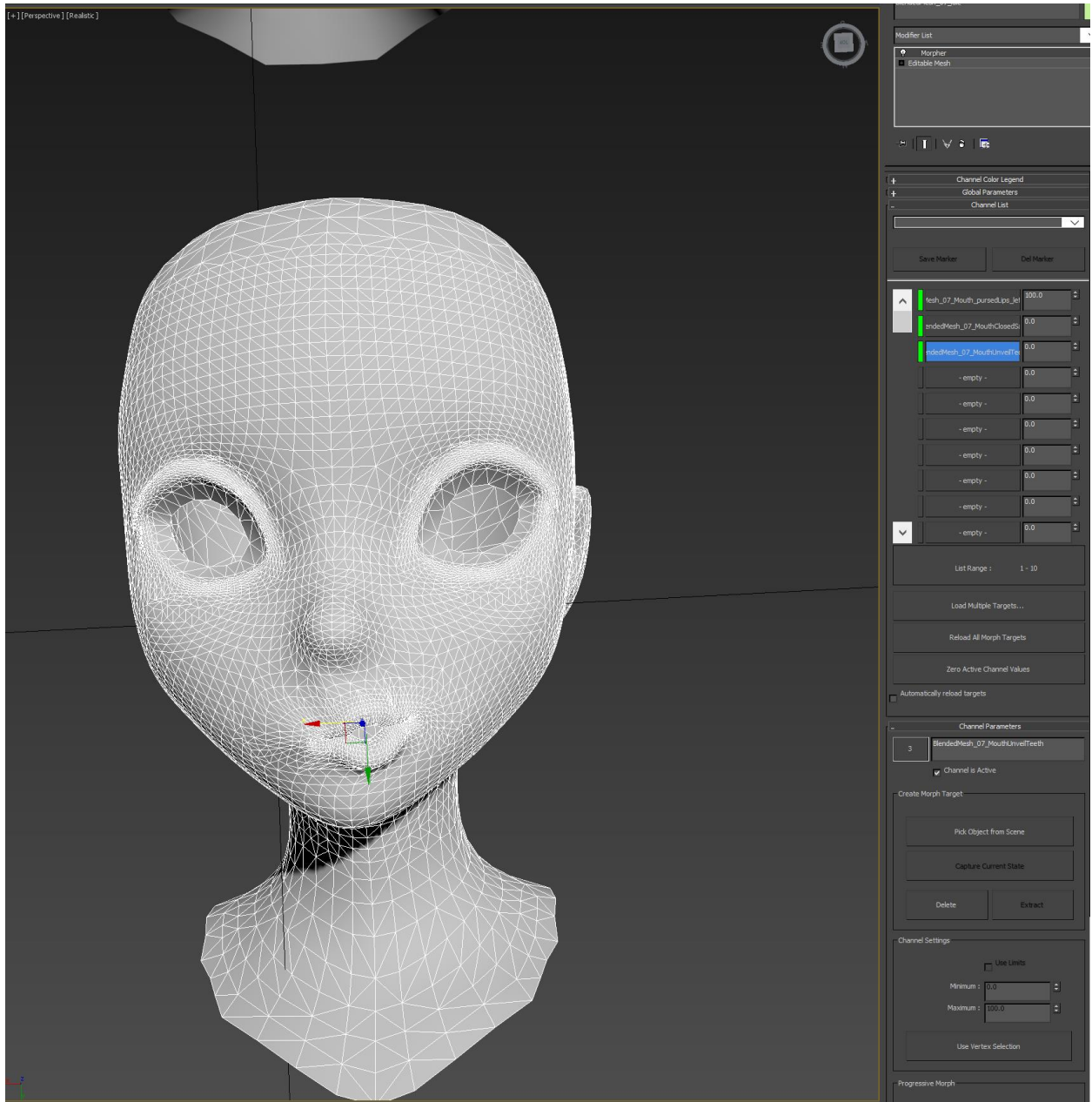


7. Scroll down and choose “Pick Object From Scene”. Now select your target object. This will become a blendshape target. Next up, select a new blendshape slot (it’s where the list of buttons named –empty – is). Then choose “Pick Object From Scene” and select your next blendshape target.

Do these steps for each blendshape target.



8. You can even test your blendshapes inside 3D Studio Max before reimporting in Unity. You just have to choose your target blendshape and change the influence procentage from 0 to 100



9. Export the above model with “Export Selected Object” into the FBX format and now import it back into Unity and you will have a model with all the blendshapes (Skinning is gone at this point). You could preserve skinning if you start with the skinned mesh, but depending on the bind pose (or T Pose) the blendshapes might not work. If you have issues try to match the bind pose with the idle frame exactly. If you want to have smooth normals for the blendshape targets, select the FBX model, go to the “Model” tab and under Normals select “Calculate”. Now

normals will be blended when blendshapes are blended.

